Alex Houle

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Education

University of New Hampshire Durham, NH

Bachelor of Science in Computer Science *Expected May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Computer Networks, Operating Systems, Systems Programming.

Experience

Test Technician March 2026 – Present

UNH Interoperability Lab *Durham, NH*

- Executed conformance test plans for Time-Sensitive Networking protocols, standards IEEE 802.1AS, 802.1Q, and 802.1BA.
- Conducted multi-vendor interoperability testing on Layer 2 networking devices, identifying configuration issues to ensure seamless communication across networking equipment.
- Performed Layer 2 MAC-level conformance testing against IEEE 802.3 standards, using packet capture tools to verify conformance.
- Performed high-speed InfiniBand interoperability testing across NICs, transceivers/DACs, and switches.

Cybersecurity Team Member Sept 2024 – Present

University Cybersecurity Club *Durham, NH*

- Competed in the NECCDC, managing Linux infrastructure, Keycloak authentication services, and Gitea repositories while completing incident response tasks and operational injects under time constraints.
- Collaborate with peers to analyze vulnerabilities, practice penetration testing, and strengthen overall security awareness.
- Hardened Linux systems during attack-and-defense cybersecurity competitions, helping maintain service availability and improve team scoring.

Projects

Chess Engine | *C, Bitboards, Systems Programming* March 2024 – October 2025

- Designed and implemented a high-performance chess engine using bitboards for efficient move generation and position evaluation.
- Implemented algorithms for move validation, game state evaluation, and search optimization (minimax, alpha-beta pruning).
- Applied low-level systems programming concepts such as memory management, bit manipulation, and Performance Optimization.

OpnLaaS (Lab as a Service) | *Go, Fiber, Proxmox API, Tailwind CSS* 2025 – 2025

- Developed a full-stack orchestration platform for automated provisioning of virtual machines and containers.
- Architected a scalable backend in Go using the Fiber framework to interface with Proxmox APIs for real-time resource management.
- Built a responsive web dashboard with Tailwind CSS, enabling users to request, monitor, and manage isolated lab environments.

Technical Skills

Languages: C, Go, Java, Python, JavaScript, HTML/CSS
Frameworks/Tools: React, Fiber, Flask, Docker, Linux, Git, Tailwind CSS

Leadership & Achievements

Eagle Scout, Boy Scouts of America

- Led a service project coordinating volunteers, resources, and logistics, demonstrating leadership, initiative, and project management skills.
- Developed resilience, teamwork, and problem-solving through years of community service and outdoor leadership.